

Unit 1: Data Representation

- **Important Topics:**
 - **Signed Number Representation** (2's complement, overflow detection)
 - **Floating Point Representation** (Fixed vs. Floating)
 - **Error Detection Codes** (Parity bit, even/odd parity)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - 2's complement arithmetic (e.g., $90 - 43$) is **asked yearly**.
 - Parity bit generation appears in short answers.

Unit 2: Register Transfer and Microoperations

- **Important Topics:**
 - **Arithmetic Microoperations** (Adder-subtractor circuit)
 - **Shift Microoperations** (Logical, arithmetic shifts)
 - **Register Transfer Language (RTL)**
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Adder-subtractor diagrams and shift operations are **frequent**.
 - RTL examples appear in short notes.

Unit 3: Basic Computer Organization and Design

- **Important Topics:**
 - **Instruction Cycle** (Fetch phase, microoperations)
 - **Common Bus System** (Diagram + explanation)
 - **Instruction Formats** (Types + symbolic representation)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Common bus system and fetch cycle are **repeated yearly**.
 - Instruction formats (e.g., zero/three-address) are tested with examples.

Unit 4: Microprogrammed Control

- **Important Topics:**
 - **Microprogram Sequencer** (Block diagram)
 - **Symbolic Microinstructions**
 - **Hardwired vs. Microprogrammed Control**
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Comparisons between hardwired and microprogrammed control are **common**.
 - Microprogram sequencer diagrams appear in long answers.

Unit 5: Central Processing Unit

- **Important Topics:**
 - **Addressing Modes** (Types + examples)
 - **RISC vs. CISC** (Pros/cons, overlapped register windows)
 - **Data Transfer Instructions**
- **Marks Weightage:** ~10 marks
- **Trends:**
 - RISC/CISC differences are **asked every year**.
 - Addressing modes (e.g., immediate, indirect) appear in short answers.

Unit 6: Pipelining

- **Important Topics:**
 - **4-Segment Pipeline** (Space-time diagram)
 - **Pipeline Hazards** (Structural, data, control)
 - **Flynn's Classification** (SIMD, MIMD)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Pipeline hazards and solutions are **frequent**.
 - Flynn's classification appears in theory questions.

Unit 7: Computer Arithmetic

- **Important Topics:**
 - **Booth's Algorithm** (Multiplication, e.g., $50 \times (-13)$)
 - **Non-Restoring Division** (Flowchart + example, e.g., $10 \div 3$)
 - **Signed Magnitude Operations**
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Booth's algorithm is **almost guaranteed**.
 - Division algorithms (restoring/non-restoring) are compared.

Unit 8: Input-Output Organization

- **Important Topics:**
 - **Priority Interrupt** (Daisy-chaining, polling)
 - **DMA Controller** (Block diagram + I/O interaction)
 - **Programmed I/O vs. Interrupt-Driven I/O**
- **Marks Weightage:** ~10 marks
- **Trends:**
 - DMA and interrupt methods are **repeated yearly**.
 - Comparisons between I/O transfer modes are common.

Unit 9: Memory Organization

- **Important Topics:**
 - **Cache Mapping** (Direct, associative)
 - **Memory Hierarchy** (Locality of reference)
 - **Associative Memory** (Match logic)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Cache mapping techniques are **frequently tested**.
 - Memory hierarchy appears in short notes.

Conclusion: Most Important Units

1. **Unit 6 (Pipelining)** and **Unit 7 (Computer Arithmetic)**: High marks (15 each), with Booth's algorithm and pipeline hazards as **must-prepare**.
2. **Unit 3 (Basic Computer Organization)**: 15 marks, focus on instruction cycle and bus system.
3. **Unit 1 (Data Representation)**: 10 marks, 2's complement and parity are **recurring**.
4. **Unit 8 (I/O Organization)**: DMA and interrupts are **frequently tested**.

Exam Strategy:

- **Priority:** Booth's algorithm, pipeline hazards, common bus system, and cache mapping.
- **Practice:** Hand-tracing division algorithms, drawing adder-subtractor circuits.
- **Short Notes:** RISC vs. CISC, addressing modes, error detection codes.

Patterns Observed:

- **Booth's Algorithm** and **Pipeline Hazards** appear **every year**.
- **Diagrams:** Common bus system, microprogram sequencer, DMA controller.
- **Comparative Questions:** Hardwired vs. microprogrammed control, RISC vs. CISC.



Unit 1: Introduction to Computer Graphics

- **Important Topics:**
 - **Graphics Hardware:** Raster vs. Vector displays, display processors
 - **Graphics Software Standards** (e.g., OpenGL)
- **Marks Weightage:** ~5-10 marks
- **Trends:**
 - Comparisons between raster/vector displays are **frequent**.
 - OpenGL basics appear in short notes.

Unit 2: Scan Conversion Algorithms

- **Important Topics:**
 - **Bresenham’s Line Algorithm** (Derivation + Tracing, e.g., endpoints (6,12) to (10,5))
 - **Mid-Point Circle/Ellipse Algorithm** (Tracing, e.g., radius=5 at (-2,-2))
 - **Fill Algorithms** (Boundary-fill vs. Flood-fill)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Bresenham’s and Mid-Point algorithms are **asked yearly**.
 - Fill algorithms tested with procedural steps.

Unit 3: 2D Geometric Transformations

- **Important Topics:**
 - **Homogeneous Transformations** (Translation, Rotation, Scaling)
 - **Composite Transformations** (e.g., rotate + translate triangle ABC)
 - **Window-to-Viewport Transformation** (Mathematical derivation)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Transformation matrices (e.g., 90° rotation + translation) are **common**.
 - Window-viewport problems appear in long answers.

Unit 4: 3D Geometric Transformations

- **Important Topics:**
 - **3D Rotation/Scaling** (Homogeneous coordinates, e.g., +30° about X-axis)
 - **Projections** (Orthographic vs. Perspective)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Orthographic/perspective comparisons are **repeated**.
 - 3D transformations often involve composite steps.

Unit 5: 3D Object Representation

- **Important Topics:**
 - **Bezier/B-spline Curves** (Properties, control points e.g., A(1,1) to D(6,4))
 - **Polygon Tables** (Surface normal, mesh representation)
 - **Wireframe & Blobby Objects**
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Bezier curves are **frequently tested** with control points.
 - Polygon tables appear in object representation questions.

Unit 6: Solid Modeling

- **Important Topics:**
 - **BSP Trees** (Recursive space subdivision)
 - **Sweep/Boundary Representations**
- **Marks Weightage:** ~5-10 marks
- **Trends:**
 - BSP tree applications in visible surface detection.

Unit 7: Visible Surface Detection

- **Important Topics:**
 - **Z-Buffer Algorithm** (Depth comparison)
 - **Painter’s Algorithm** (Depth sorting)
 - **Scan-Line Method** (vs. Depth Buffer)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Z-buffer and Painter’s algorithms are **compared yearly**.
 - Scan-line filling tested with polygon examples.

Unit 8: Illumination Models

- **Important Topics:**
 - **Phong Shading** (Specular reflection, advantages)
 - **Gouraud vs. Phong**
 - **Intensity Attenuation** (Shadows)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Phong model is **almost guaranteed**.
 - Gouraud shading appears in short comparisons.

Unit 9: Virtual Reality

- **Important Topics:**
 - **VR vs. AR** (Examples, components)
 - **Applications** (e.g., navigation interfaces)
- **Marks Weightage:** ~5 marks
- **Trends:**
 - VR/AR differences are **common** in short notes.

Unit 10: OpenGL

- **Important Topics:**
 - **Basic Commands** (Drawing pixels, polygons)
 - **Callback Functions**
- **Marks Weightage:** ~5 marks
- **Trends:**
 - OpenGL syntax appears in programming-based questions.

Conclusion: Most Important Units

1. **Unit 2 (Scan Conversion)** and **Unit 7 (Visible Surface Detection)**: High marks (15 each), with Bresenham’s/Z-buffer as **must-prepare**.
2. **Unit 5 (3D Object Representation)**: 15 marks, focus on Bezier curves and polygon tables.
3. **Unit 3 (2D Transformations)**: 10 marks, composite transformations are **recurring**.
4. **Unit 8 (Illumination)**: Phong shading is **frequently tested**.

Exam Strategy:

- **Priority:** Bresenham’s algorithm, Z-buffer, Bezier curves, and homogeneous transformations.
- **Practice:** Tracing algorithms (Mid-Point, Bresenham), Phong shading steps.
- **Short Notes:** VR vs. AR, OpenGL commands, fill algorithms.

Patterns Observed:

- **Bresenham’s Algorithm** and **Z-Buffer** appear **every year**.
- **Diagrams:** Mid-Point circle, window-viewport transformation.
- **Comparative Questions:** Gouraud vs. Phong, raster vs. vector.

Final Note: Focus on **algorithm derivations**, **3D transformations**, and **illumination models** for comprehensive coverage.

Unit 1: Introduction

- **Important Topics:**
 - **Asymptotic Notations** (Big-O, Theta, Omega)
 - **Abstract Data Type (ADT)** vs. Data Structure
 - Dynamic Memory Allocation in C
- **Marks Weightage:** ~5-10 marks
- **Trends:**
 - Big-O notation is **asked every year** (e.g., find Big-O of $5x^4 + 9x^2 + 7x + 9$).
 - ADT definitions appear in short notes.

Unit 2: Stack

- **Important Topics:**
 - **Infix to Postfix Conversion** (e.g., $A + (B - C) * (D - E) / G$)
 - **Stack ADT** (Operations: Push/Pop)
 - Applications (Expression evaluation, recursion stack)
- **Marks Weightage:** ~10-15 marks
- **Trends:**
 - Infix-postfix conversion is **repeated yearly**.
 - Stack implementation (array/linked list) appears in programming questions.

Unit 3: Queue

- **Important Topics:**
 - **Circular Queue** (Implementation + Need)
 - **Queue ADT** (Enqueue/Dequeue)
 - Priority Queue (Short notes)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Circular vs. Linear Queue comparisons are frequent.
 - Queue applications (e.g., BFS) linked with Graph Theory.

Unit 4: Recursion

- **Important Topics:**
 - **Tail Recursion** (vs. Iteration)
 - **Recursive Algorithms** (GCD, Fibonacci, TOH)
- **Marks Weightage:** ~5-10 marks
- **Trends:**
 - GCD/Fibonacci programs are **common**.
 - Tower of Hanoi (TOH) appears in short notes.

Unit 5: Lists

- **Important Topics:**
 - **Doubly/Circular Linked Lists** (Insert/Delete at k-th position)
 - **Stack/Queue using Linked List**
 - Array vs. Linked List comparisons
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Linked list operations are **asked every year**.
 - Doubly linked list questions often include diagrams.

Unit 6: Sorting

- **Important Topics:**
 - **Divide-and-Conquer Sorts** (Quick, Merge, Heap)
 - **Comparison Sorts** (Bubble, Selection, Insertion)
 - Shell Sort (Tracing steps)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Quick Sort (pivot analysis) and Merge Sort are **frequent**.
 - Hand-tracing sorting steps (e.g., Shell Sort) is common.

Unit 7: Searching and Hashing

- **Important Topics:**
 - **Binary Search** (Algorithm + Time Complexity)
 - **Hashing** (Linear/Quadratic Probing, Rehashing)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Binary search is **almost guaranteed**.
 - Collision resolution techniques (e.g., $h(x) = x \% 5$) are repeated.

Unit 8: Trees and Graphs

- **Important Topics:**
 - **AVL Trees** (Construction + Balancing)
 - **BST Operations** (Insert/Delete)
 - **Graph Algorithms** (Dijkstra, Kruskal/Prim's MST)
- **Marks Weightage:** ~20-25 marks
- **Trends:**
 - AVL tree construction (e.g., input: 24,12,8,...) is **asked yearly**.
 - Dijkstra's algorithm appears in long answers.

Conclusion: Most Important Units

1. **Unit 8 (Trees/Graphs)** and **Unit 6 (Sorting)**: High marks (20+ each), with AVL trees and sorting algorithms as **must-prepare**.
2. **Unit 2 (Stack)** and **Unit 5 (Lists)**: 10-15 marks, focus on conversions and linked list operations.
3. **Unit 7 (Searching/Hashing)**: Binary search and hashing are **recurring**.
4. **Unit 1 (Introduction)**: Asymptotic notations are **frequently tested**.

Exam Strategy:

- **Priority:** AVL trees, sorting algorithms, stack conversions, and linked lists.
- **Practice:** Hand-tracing sorts, infix-postfix conversions, and recursive algorithms.
- **Short Notes:** Tail recursion, collision resolution, and ADT definitions.

Patterns Observed:

- **AVL Trees** and **Binary Search** appear **every year**.
- **Programming Questions:** Linked list operations, recursive GCD/Fibonacci.
- **Comparative Questions:** Stack vs. Queue, Array vs. Linked List.

Chapter 1: Errors and Mathematical Review

- **Important Topics:**
 - Types of Errors (True, Relative, Approximate)
 - Propagation of Errors
 - Taylor Theorem and Error in Taylor Series
- **Marks Weightage:** ~5-10 marks (Often combined with other chapters)
- **Trends:**
 - Definitions and error types are frequently asked in short questions.
 - Taylor series applications appear occasionally in derivations.

Chapter 2: Solving Nonlinear Equations

- **Important Topics:**
 - **Newton-Raphson Method** (Derivation + Solving, e.g., $x^4 - x - 10 = 0$)
 - **Secant Method** (Derivation + Solving, e.g., $\cos x + 2 \sin x - x^2 = 0$)
 - Bisection Method (Real root estimation, e.g., $x^2 - e^{-x} = 3$)
 - Fixed-Point Iteration (Convergence conditions)
- **Marks Weightage:** ~15-20 marks (2-3 questions per exam)
- **Trends:**
 - Newton-Raphson and Secant methods are **almost every year**.
 - Bisection/Fixed-Point asked for specific precision requirements.

Chapter 3: Interpolation

- **Important Topics:**
 - **Lagrange Interpolation** (Algorithm + Program + Solving, e.g., estimate $y(10)$)
 - **Newton’s Divided Difference** (Table construction + Estimation, e.g., missing term)
 - Cubic Spline (Natural spline conditions + Estimation, e.g., $f(2)$)
- **Marks Weightage:** ~10-15 marks
- **Trends:**
 - Lagrange and Newton’s interpolation are **repeated yearly**.
 - Spline interpolation appears in 50% of exams.

Chapter 4: Regression Analysis

- **Important Topics:**
 - **Least Squares Linear Regression** (Fit $y = ax + b$, e.g., given tabular data)
 - Polynomial Regression (Quadratic fit, e.g., $y = ax^2 + bx + c$)
 - Exponential Fit (Linearization, e.g., $y = ae^{bx}$)
- **Marks Weightage:** ~5-10 marks
- **Trends:**
 - Linear regression is common; exponential fit appears occasionally.

Chapter 5: System of Linear Equations

- **Important Topics:**
 - **Gauss-Jordan Elimination** (Solving + Matrix Inversion)
 - **LU Decomposition** (Doolittle/Cholesky, e.g., factorize given matrix)
 - **Iterative Methods** (Jacobi vs. Gauss-Seidel, e.g., solve $2x - 7y - 10z = 17$)
 - Ill-conditioning and Pivoting
- **Marks Weightage:** ~15-20 marks
- **Trends:**
 - Direct methods (Gauss-Jordan/LU) and iterative methods are **frequent**.
 - Ill-conditioning discussed in theory questions.

Chapter 6: Numerical Differentiation

- **Important Topics:**
 - Three-Point Formula (Derivatives for tabulated data, e.g., velocity/acceleration)
 - Maxima/Minima of Tabulated Functions
- **Marks Weightage:** ~5 marks
- **Trends:**
 - Practical applications (e.g., estimating velocity) appear sporadically.

Chapter 7: Numerical Integration

- **Important Topics:**
 - **Simpson’s 1/3 Rule** (Derivation + Solving, e.g., $\int \frac{e^x}{x} dx$)
 - **Trapezoidal Rule** (Composite form, e.g., 2/4 segments)
 - Romberg Integration (e.g., $\int \frac{\sin x}{x} dx$)
- **Marks Weightage:** ~10-15 marks
- **Trends:**
 - Simpson’s rule is **asked every year**; Romberg occasionally.

Chapter 8: Ordinary Differential Equations

- **Important Topics:**
 - **Runge-Kutta 4th Order** (Solving IVP, e.g., $\frac{dy}{dx} = \frac{x}{y}, y(0) = 1$)
 - Euler’s/Heun’s Method (Algorithm + Program)
 - Shooting Method (BVP, e.g., $\frac{d^2y}{dx^2} = 6x, y(1) = 2, y(2) = 9$)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - RK4 and Euler’s methods are **repeated yearly**; Shooting method in 30% exams.

Chapter 9: Partial Differential Equations

- **Important Topics:**
 - **Poisson’s Equation** (Finite difference, e.g., $\nabla^2 u = -64xy$)
 - Laplace’s Equation (Steady-state heat distribution, e.g., plate temperatures)
- **Marks Weightage:** ~10 marks
- **Trends:**
 - Poisson’s equation is common; applications in boundary conditions.

Conclusion: Most Important Chapters

1. **Chapter 2 (Nonlinear Equations)** and **Chapter 5 (Linear Systems)**: High marks (15-20 each), frequent derivations + numerical problems.
2. **Chapter 3 (Interpolation)** and **Chapter 8 (ODEs)**: 10-15 marks, algorithms + applications.
3. **Chapter 7 (Integration)**: Simpson’s rule is almost guaranteed.
4. **Chapter 9 (PDEs)**: Poisson’s equation is recurring.

Exam Strategy: Focus on Newton-Raphson, LU Decomposition, Lagrange/RK4, and Simpson’s rule for guaranteed marks. Practice derivations and algorithms for interpolation/ODEs.

Unit 1: Sampling Distribution and Estimation

- **Important Topics:**
 - **Central Limit Theorem (CLT)** (Numerical applications, e.g., sample mean distribution)
 - **Confidence Intervals** (Mean, proportion; e.g., 96% CI for toothpaste leakage)
 - **Sample Size Determination** (Linked to error margin, e.g., CV=60%)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - CLT is **tested yearly** with practical implications (e.g., bulb lifespan).
 - Confidence intervals for proportions (e.g., defective chips) are common.

Unit 2: Testing of Hypothesis

- **Important Topics:**
 - **t-tests** (One-sample, two-sample, paired; e.g., reaction time before/after drug)
 - **p-value Interpretation** (Decision-making at $\alpha=0.05/0.01$)
 - **Proportion Tests** (e.g., Samsung vs. Redmi mobile popularity)
- **Marks Weightage:** ~20 marks
- **Trends:**
 - Paired t-tests (e.g., left/right hand strength) and proportion tests are **frequent**.
 - Linkage between CI and hypothesis testing is often asked.

Unit 3: Non-Parametric Tests

- **Important Topics:**
 - **Kruskal-Wallis Test** (e.g., missile system burning rates)
 - **Wilcoxon-Mann-Whitney** (e.g., IT worker knowledge scores)
 - **Chi-square Test** (Independence, e.g., email hacking vs. account type)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Kruskal-Wallis and Chi-square tests appear **almost every year**.
 - Run test for randomness (e.g., coin toss sequence) is recurring.

Unit 4: Multiple Correlation and Regression

- **Important Topics:**
 - **Multiple Regression** (Equation derivation, e.g., unpaid taxes vs. audit hours)
 - **Partial/Multiple Correlation** (e.g., $R_{1.23}$ with given r_{12}, r_{13}, r_{23})
 - **Model Adequacy** (Adjusted R^2 , multicollinearity)
- **Marks Weightage:** ~20 marks
- **Trends:**
 - Regression coefficient interpretation (e.g., job satisfaction vs. aptitude/absenteeism) is **key**.
 - Partial correlation calculations are frequently tested.

Unit 5: Design of Experiment

- **Important Topics:**
 - **ANOVA** (CRD, RBD, LSD; e.g., yield comparisons in LSD)
 - **Missing Value Estimation** (LSD/RBD; e.g., 4x4 experiment)
 - **Efficiency Comparisons** (RBD vs. CRD)
- **Marks Weightage:** ~25 marks
- **Trends:**
 - LSD analysis (e.g., 3x3 layout) is **asked yearly**.
 - ANOVA tables (e.g., incomplete data) require hands-on practice.

Unit 6: Stochastic Process

- **Important Topics:**
 - **Markov Chains** (Steady-state, e.g., sunny/rainy day forecasts)
 - **Queuing Theory** (M/M/1, Little's law; e.g., printer message system)
 - **Poisson Process** (e.g., job arrivals in networks)
- **Marks Weightage:** ~15 marks
- **Trends:**
 - Markov chains (transition matrices) and queuing metrics (wait time) are **common**.

Conclusion: Most Important Units

1. **Unit 5 (Design of Experiment) and Unit 4 (Regression):** High marks (20-25 each), with ANOVA and regression as **core topics**.
2. **Unit 2 (Hypothesis Testing):** 20 marks, focus on t-tests and p-values.
3. **Unit 3 (Non-Parametric Tests):** 15 marks, Kruskal-Wallis/Chi-square are **repeated**.
4. **Unit 6 (Stochastic):** Markov chains and queuing theory are **frequently tested**.

Exam Strategy:

- **Priority:** ANOVA (LSD), multiple regression, and non-parametric tests.
- **Practice:** Derive regression equations, complete ANOVA tables, solve Markov chain problems.
- **Short Notes:** CLT, queuing components, Adjusted R^2 .

Patterns Observed:

- **ANOVA** and **Regression** dominate long-answer questions.
- **Non-parametric tests** (e.g., Kruskal-Wallis) appear in both theory and numericals.
- **Stochastic processes** often include real-world scenarios (e.g., weather forecasts).

Final Note: Master **ANOVA designs, regression interpretation, and Markov chains** for comprehensive coverage. Practice **missing value estimation** and **queuing calculations** for problem-solving sections.

 This response is AI-generated, for reference only.